Journal of Food Composition and Analysis 135 (2024) 106593

Contents lists available at ScienceDirect

Journal of Food Composition and Analysis

journal homepage: www.elsevier.com/locate/jfca

Combination of gas chromatography-mass spectrometry and hyperspectral imaging for identification of adulterated Safflower seed oil

Zhiyong Zou [a,1], Qianlong Wang [a,1], Menghua Li [a], Jiangbo Zhen [a], Dongyu Yuan [a], Qing Zhang [b], Qiang Cui [b], Meiliang Li [b], Man Zhou [b], Shutao Yin [c,*], Lijia Xu [a,*]

[a] College of Mechanical and Electrical Engineering, Sichuan Agricultural University, Ya'an 625014, China
[b] College of Food Science, Sichuan Agricultural University, Ya'an 625014, China
[c] Institute of Modern Agricultural Industry, China Agricultural University, Chengdu, Sichuan 611430, China

ARTICLE INFO

Keywords:
Hyperspectral imaging technology
Safflower seed oil
Fatty acid
Adulteration concentration
Machine learning

ABSTRACT

Safflower seed oil (SSO), a crucial edible oil, occupies a significant position in human diet due to its unique nutritional components. However, because of its high economic value, SSO has become a primary target for food adulteration. To rapidly and effectively detect the concentration of adulterants in SSO, a research approach using hyperspectral imaging and gas chromatography-mass spectrometry (GC-MS) was proposed as technical means, employing machine learning as the methodology. Specifically, different preprocessing methods were compared, with median filtering (MF) selected for spectral data to significantly reduce noise and improve the robustness and generalization ability of the model. Regarding the selection of feature bands, bands around 440 nm, 530 nm, and 880 nm – 950 nm were identified as more favorable for establishing a predictive model for the concentration of adulterants in SSO, while also shortening the modeling time. By constructing an ensemble learning model with ridge regression (Ridge) and partial least squares regression (PLSR) as base models and LightGBM as the meta-model, achieving high-precision prediction of the concentration of adulterants in SSO. Furthermore, by jointly modeling the linoleic acid, oleic acid and palmitic acid measured by GC-MS with hyperspectral data, the model's R^2 was improved to 0.976, highlighting its outstanding performance. Therefore, this study identifies MF-Ridge-Stacking as the optimal model for predicting the concentration of adulterants in SSO. The research not only provides theoretical and technical support for adulteration identification but also presents a new method for predicting the concentration of adulterants in SSO, with potential practical application value.

1. Introduction

Safflower seed oil (SSO) is particularly rich in unsaturated fatty acid (Khalid et al., 2017). Among the known vegetable oils, it has the highest mass fraction of linoleic acid, reaching up to 85 % (Kammili and Yadav, 2022). SSO has many benefits such as antioxidants, which promotes blood circulation and improves the function of various organs in the body, thereby delaying aging (Fadiloglu et al., 2023). However, due to its outstanding properties, the price of SSO is often higher than that of other vegetable oils, making it a primary target for economically motivated adulteration and fraud (Uncu et al., 2019). The most typical fraud involving SSO is the adulteration by blending other inexpensive vegetable oils such as sunflower oil (SFO) and soybean oil (SO) into SSO (Han et al., 2022). Such adulteration practices may lead consumers to unknowingly consume the adulterated product, posing risks to human health and disrupting market order (Kanny et al., 1994).

Faced with the varying quality of the market, numerous researchers have proposed several vegetable oil identification methods, such as chromatography (Hertzog et al., 2022), mass spectrometry (Cao et al., 2021; S.-L. Wang et al., 2021), UV–visible spectrophotometry (Johnson et al., 2022) and triacylglycerol pseudo-targeting method (Zhang et al., 2023). However, while these methods are widely adopted, they often require complex sample pretreatment and analysis procedures. Additionally, they may not perform well in distinguishing adulterated oils with similar fatty acid contents. Furthermore, these procedures are time-consuming, labor-intensive, expensive, and do not meet the requirements for rapid analysis of adulterated oil products. Therefore, some rapid and non-destructive detection methods are also widely

* Corresponding authors.
E-mail addresses: yinshutao@cau.edu.cn (S. Yin), xulijia@sicau.edu.cn (L. Xu).
[1] These authors contributed equally to this work and should be considered co-first authors.

https://doi.org/10.1016/j.jfca.2024.106593
Received 30 January 2024; Received in revised form 20 June 2024; Accepted 28 July 2024
Available online 2 August 2024
0889-1575/© 2024 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

applied in oil identification.

Currently, spectroscopic technology has been used to identify oils. For instance, nuclear magnetic resonance (NMR) technology is employed to distinguish coconut oil (Bao et al., 2023), Raman spectroscopy for identifying waste cooking oil (Jin et al., 2021), fluorescence hyperspectroscopy for rapid identification of adulterated sesame oil (Pan et al., 2021), and terahertz spectroscopy for differentiating crude oils from various fields (Zhan et al., 2015). Compared to physical and chemical methods, spectroscopic analysis offers advantages such as speed, ease of use, simplicity, minimal sample preparation, and non-destructiveness (Lewis, 2023). However, despite NMR's high resolution and sensitivity to structure, its signal intensity may be insufficient for highly diluted samples or molecules with low concentrations, leading to a low signal-to-noise ratio that compromises data quality and reliability (Syafri et al., 2022). Meanwhile, despite their high sensitivity, Raman spectroscopy and fluorescence hyperspectroscopy can suffer from reduced identification accuracy due to strong fluorescent groups (Ríos-Reina et al., 2024). Terahertz spectroscopy is easily affected by the environment, leading to instability in system operation (Hou et al., 2024). Fortunately, hyperspectral imaging (HSI) technology has emerged as a widely adopted method for quality assessment of edible oils, addressing limitations of traditional physical and chemical analyses (Jermwongruttanachai et al., 2023). This technology can record the overtone and combined band information of the basic vibratio n of a single chemical bond in the molecule, and can provide structural information between different functional groups in SSO (Meng et al., 2023). The chemical bonds in SSO, such as C-H, O-H, ester bonds, and carbon-carbon double bonds, provide unique opportunities for this purpose. Therefore, HSI as a rapid and non-destructive detection method, holds significant importance in the prediction of adulteration and component analysis of SSO.

HSI technology possesses the capability to handle large-scale samples and generate complex multidimensional data, while GC-MS is effective in measuring the fatty acid content information of SSO. Considering that chromatographic information is also a collection of multidimensional data, several studies highlight the necessity of using chemometric approaches to enhance the interpretation and optimization of these complex results (Contreras et al., 2019; Gerhardt et al., 2019; Jurado-Campos et al., 2020). In this study, a comprehensive approach involving HSI technology, machine learning, and GC-MS was adopted to compare the adulteration of different oil varieties with SSO. Specifically, (1) through comparative analysis, an effective data preprocessing method was identified, providing a robust foundation for subsequent spectral studies. (2) In the investigation of adulteration in SSO, relevant feature bands was delved into in order to find the optimal predictive model for achieving rapid quantitative analysis of adulteration. (3) By comparing the performance of ensemble learning models with traditional machine learning models, the advantages of ensemble learning models were revealed in adulteration studies, offering valuable insights for model selection in related fields in the future. (4) Simultaneously, physicochemical values as feature variables were incorporated to explore ways to enhance predictive accuracy, providing new perspectives for further optimizing predictive models.

2. Materials and methods

The overall study is shown in Fig. 1. A hyperspectrometer (Fig. 1a) was used to collect spectral data of oil samples, and the content and concentration of linoleic acid (LA), oleic acid (OA) and palmitic acid (PA) in the oil samples were detected by GC-MS (Fig. 1b). Coupled with machine learning, quantitative analysis of SSO adulteration in oil samples was conducted to explore the internal relationship between spectral data and fatty acid content.

2.1. Oil samples

This study focused on using first-grade cold-pressed SSO, processed from specially selected imported raw materials from Kazakhstan

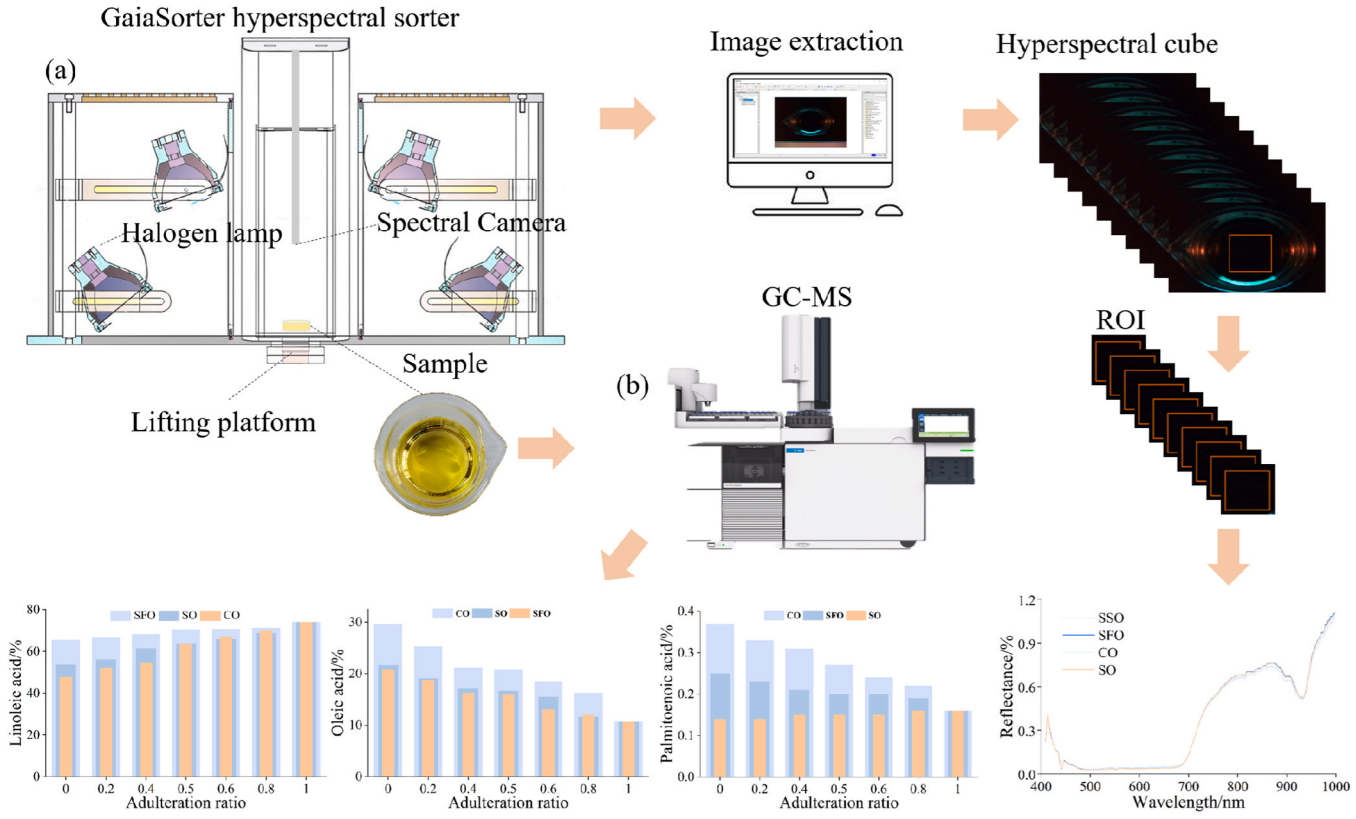


Fig. 1. Overview of the overall research route.

(COFCO Tayuan Safflower (Xinjiang) Co., Ltd.). Vegetable oils from well-known Chinese brands were purchased from a local large supermarket in Ya'an, China. These included SFO, SO, and corn oil (CO), ensuring the reliability and universality of the research results. All selected oil samples met the requirements of the Chinese National Standards for Edible Vegetable Oils (GB 2716-2018). The selection of oils from different sources was based on their representativeness and quality assurance, providing extensive data support for the study and ensuring the credibility of the experimental results. Table 1 provides detailed information on the blending ratios of SFO, SO, and CO with various volume fractions of SSO. Prior to mixing, pure oil samples were homogenized. A total of 350 samples (50 parts of pure oil and 300 parts of adulterated oil) were prepared and analyzed. Before spectroscopic analysis, the oil blends were uniformly mixed using a vortex mixer and stored in a dark environment at 4°C for further analysis.

2.2. HSI and data acquisition

The data for this study were collected at the Hyperspectral Laboratory of Sichuan Agricultural University, the GaiaSorter hyperspectral photometer (ZOLIX INSTRUMENTS CO., LTD, China) was used to obtain the spectral information of the prepared oil samples. The system's core components comprised a hyperspectral camera, a light source, a motorized stage, a computer, and control software. The effective spectral range was 400–1000 nm, with a spectral resolution of 2.8 nm and a total of 256 bands. The distance between the hyperspectral camera lens and the oil sample loading platform was set at 160 mm, with a motorized stage speed of 4.6 mm/s, and an exposure time of 8.5 ms for the hyperspectral camera. Spectral images for each oil sample were collected individually, with three scans per oil sample used to calculate the average spectrum. A total of 350 samples were scanned, resulting in 89,600 spectral data values. To eliminate the influence of dark current in the hyperspectral camera and enhance the accuracy of spectral data acquisition, the hyperspectral equipment was preheated for half an hour. Black and white calibration was also performed using the Specview software, following the calibration formula (Wang and Song, 2023):

$$R = \frac{I - B}{W - B} \tag{1}$$

Where R, I, W, and B respectively represent the calibrated data, white frame data, black frame data, and original spectral data.

2.3. Content measurement of linoleic acid, oleic acid and palmitic acid

The experiments were conducted using an Agilent 7890A-5975 C GC-MS system manufactured by Agilent Technologies, USA. During the analysis of each sample, 200–300 mg of the mixed oil sample was placed in a 15 mL vial with a stopper. Subsequently, 1 mL of anhydrous toluene and 1 mL of a 0.5 mol/L sodium methoxide-methanol solution were added (Liu et al., 2020). After shaking the mixture thoroughly, it was placed in a 50°C oven for a 10-min incubation. Upon cooling to room temperature, 0.1 mL of glacial acetic acid and 1 mL of deionized water were added. After thorough mixing, extraction was performed using 5 mL of n-hexane. After allowing the layers to separate, 1 mL of the supernatant was collected, and further diluted 5 times with n-hexane (Spadafora et al., 2022). Finally, the samples were dehydrated using anhydrous sodium sulfate and filtered through a 0.22 μm organic membrane. The processed samples were stored at −4°C until analysis. During the data acquisition phase, gas chromatography conditions involved the detection and analysis of fatty acid methyl esters on GC-MS after derivatization. To reduce measurement errors, the average values of LA, OA, and PA content for each sample were used as the final values for that particular oil sample. In this process, 1 μL of the derivative n-hexane solution was injected into the inlet at 250°C, and the carrier gas (99 % helium) flowed through the HP-5MS capillary column (30 m×0.25 mm×0.25 μm) at a constant rate of 1 mL/min without splitting. The temperature program was set as follows: the initial temperature of the column oven was 120°C, stabilized for 2 min, then ramped at a rate of 4 °C/min to 200°C and stabilized for 2 min, finally ramped at a rate of 3 °C/min to 240°C and stabilized for 2 min. In terms of mass spectrometry, full-scan mode was employed for signal acquisition with a mass-to-charge ratio (m/z) scanning range of 40–400 amu. The temperatures for the ion source and quadrupole were set at 230°C and 150°C, respectively.

Table 1
Adulterated SSO samples.

Group	A Adulterated with B (A+B)	Adulteration Ratio (%) (V_B/V_{A+B})	Quantity per category
Adulterated oil	SSO+SFO	20, 40, 50, 60, 80	20
	SSO+SO		
	SSO+CO		
Pure oil	CO	100	10
	SO	100	
	SFO	100	
	SSO	0	20

SSO – safflower seed oil; SFO – sunflower oil; SO – soybean oil; CO – corn oil.

2.4. Evaluation indicators

The evaluation criteria of the regression model are three indicators: (R^2), root mean square error (RMSE) and fitting time. The closer the value is to 1, the more it represents the predicted value. The greater the accuracy, the lower the value, indicating reduced prediction error and improved model effectiveness. 5-fold cross-validation is used to test the model. The evaluation indicators are cross-validation coefficient of determination (R^2_{cv}) and cross-validation root mean square error ($RMSE_{cv}$). The calculation formulas for the R^2 and RMSE are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (x_i - y_i)^2}{\sum_{i=1}^{n} (x_i - \overline{x})^2} \tag{2}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (x_i - y_i)^2}{n}} \tag{3}$$

In the formula, x_i represents the measured value of the i-th sample; y_i represents the true value of the i-th sample; $\overline{x}$ represents the average of the measured values of all samples; n represents the number of samples.

2.5. Extraction and preprocessing of spectral data

The experiment used the software ENVI 5.3 (Exelis Visual Information Solutions Inc., USA) (Tian et al., 2023) to intercept the spectral data of the area of interest in the corrected oil sample spectral image. Effective preprocessing can eliminate the influence of environmental factors and non-quality factor information of the spectral equipment itself in the collected hyperspectral data, and obtain more valuable spectral data. Three spectral preprocessing methods were employed in the study to enhance the signal-to-noise ratio of the black-and-white corrected spectral data, namely L2 norm normalization (L2NN) (Aulia et al., 2022), multiplicative scatter correction (MSC) (Cui et al., 2022), median filtering (MF) (Appiah et al., 2022). These three preprocessing methods allow the data to eliminate the interference of non-quality factor information on subsequent data processing to varying degrees. As shown in Fig. S1, compared with the original spectrum, the difference is that the spectral curves after L2NN are more dispersed in the range of 700 – 1000 nm; After MSC, the overall separation between spectral curves intensifies; after MF, the overall distance between spectral curves

becomes more contracted.

2.6. Regression models

Ridge regression (Ridge) is a biased estimation method used for collinear data analysis and is an improved least squares method. It abandons the unbiasedness of least squares and builds stronger tolerance for biased data, so it can be applied to regressions that better match actual situations (Fujioka and Sun, 2022). According to the Gauss model, the estimation of β can be done by Ridge regression, which is an estimation method that introduces regularization terms on the basis of ordinary least squares to solve the problem of multicollinearity. It is expressed by formula (4):

$$\beta^{k} = (X^{T}X + kI)^{-1}X^{T}Y \tag{4}$$

LightGBM is a gradient boosting algorithm framework based on the decision tree algorithm. In this algorithm, GB denotes Gradient Boosting Machine, specifically referring to gradient boosting trees, which belong to the Boosting algorithm category (Chen et al., 2019). LightGBM can be trained sequentially by fitting the negative gradient of the loss function. According to Eq. (5), the LightGBM model $F_M(x)$ can be obtained through the weighted combination scheme.

$$F_M(x) = \sum_{m=1}^{M} y_m h_m(x) \tag{5}$$

In the formula, M is the maximum number of iterations and $h_m(x)$ is the basic decision tree.

Random Forest (RF) is a supervised ensemble learning algorithm that employs attribute random selection methods to build multiple decision trees, and finally combines the prediction results of each classification tree to compare the prediction results (Wang et al., 2018). In RF, each tree is trained using bootstrapped resampling, and the training of the leaf nodes of each tree involves a certain proportion of non-replacement random sampling. As bootstrapped resampling ensures that some data are not repeatedly sampled in each iteration, multiple sets with different classification results are obtained. A decision tree is built for each set, resulting in numerous independent decision trees. The classification error depends on the classification ability of each tree and their correlation. The final prediction of the RF is obtained by summing and averaging the results of all decision trees (Abellán et al., 2018).

Gradient Boosting Decision Trees (GBDT) is an iterative decision tree algorithm composed of multiple decision trees. The conclusions of all trees are accumulated to form the final answer. The GBDT algorithm is convergent, and it can continually improve the accuracy of the final classifier by reducing bias (Liu et al., 2021).

Suppose the existence of K decision trees,

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i), f_k \in F \tag{6}$$

In formula (6), F is the function space that accommodates the entire set of regression trees, $f_k(x_i)$ represents a tree.

The goal of GBDT is:

$$O_{bj} = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{7}$$

In formula (7), $\sum_{i=1}^{n} l(y_i, \hat{y}_i)$ is the training lost, $\sum_{k=1}^{K} \Omega(f_k)$ is the complexity of the tree.

Catboost is a machine learning algorithm based on gradient boosting decision trees, which is different from other gradient boosting algorithms. Catboost adopts a symmetric tree structure, which helps avoid overfit, enhances reliability, and speeds up prediction time (Ding et al., 2021). During the tree building process of Catboost, each tree is built based on the residual of the previous tree. This iterative process makes the final prediction result more accurate and the model more robust.

Partial least squares regression (PLSR) was initially proposed by S. Wold and C. Alano in 1983. The first component t_1 was first proposed in the independent variable set (t_1 is a Linear combination of $\{x_1, x_2, x_3 \ldots x_m\}$ and extract as much variation information from the original independent variable set as possible). At the same time, the first component u_1 is also extracted from the dependent variable set, and the correlation between t_1 and u_1 is required to reach the maximum (Saha et al., 2023). Then establish a regression between the dependent variables $y_1, y_2, y_3 \ldots y_p$ *and* t_1. If the regression equation has reached satisfactory accuracy, the algorithm is terminated. Otherwise, continue to extract the second pair of components until satisfactory accuracy is achieved (Zhou et al., 2022). If r components $t_1, t_2, t_3 \ldots t_r$ are finally extracted from the independent variable set, the PLSR will establish the regression formula of $y_1, y_2, y_3 \ldots y_p$ and $t_1, t_2, t_3 \ldots t_r$, and then express it as $y_1, y_2, y_3 \ldots y_p$ and the regression equation of the original independent variables, which is the PLSR equation (Wei et al., 2022).

Stacking is a type of "integrated learning" method (Zhao and Cheng, 2022), as shown in Fig. 2a. It combines predictions from multiple base models with meta-models. Divide all data into D_{train} and D_{test}, the former is used to train the Stacking model, and the latter is used for testing. Training folds and Validation folds are also divided from D_{train}. The former is used to train the primary learner, and the latter obtains the prediction results through the primary trainer. The final prediction results are sent to the secondary learner to calculate the weight of each primary learner. N learners can be used to process the sample, and the processing results are used as the input of the next layer of learners, and terminate at the number of specified iterations.

3. Results and discussion

3.1. Fatty acid content and spectral properties of SSO and other vegetable oils

Fig. 3 presents the fatty acid content and average spectral curves of mixed oil samples at different adulteration concentrations. As can be seen from Fig. 3(a-c), there are obvious differences in the contents of LA, OA, and PA. Specifically, SSO shows the highest LA content, the lowest OA content, and relatively low PA content, forming its unique characteristics; the LA, OA, and PA contents in SFO are all medium, with no outstanding points; CO has the lowest LA content, and the highest OA and PA contents; SO has low LA and PA contents, and medium OA content. SFO and SO show similar fatty acid compositions overall, but there is a certain gap in PA (Peng et al., 2015). At the same time, it is obvious that SSO is rich in LA.

In terms of spectral data processing, serious interference bands before 400 nm and after 1000 nm are removed, and spectral data from 400 – 1000 nm are mainly analyzed. Fig. 3(d-f) illustrates the average spectra of all mixed oil samples collected using the hyperspectral imager. SSO and the other three vegetable oils exhibit different spectral intensity reflectance changes at different adulteration concentrations. The spectral region of 525 nm – 670 nm is related to the presence of vitamin E and chlorophyll in the oil, as these substances usually exhibit specific absorption properties in this wavelength range (Mburu et al., 2021). Furthermore, the spectral trends presented in the vicinity at 920 nm – 1000 nm can be attributed to the stretching and deformation vibrations of the -CH bonds, which are often associated with specific chemical bonds present in oils (Cruz-Tirado et al., 2023). Within this range, CO demonstrates stronger reflectivity, likely due to its high oleic acid content. Overall, pure SSO has the lowest spectral reflectance. Especially in the 490 nm - 530 nm and 880 nm – 950 nm bands, the difference between SSO and the other three vegetable oils is particularly significant. Generally speaking, each vegetable oil has its own unique fatty acid composition (Spencer et al., 1976). The spectral curve of SSO is unique among several vegetable oils, perhaps due to its unique LA content and trace amounts of antioxidant substances (Rudolphi et al., 2012). These differences in fatty acid content and spectral profiles

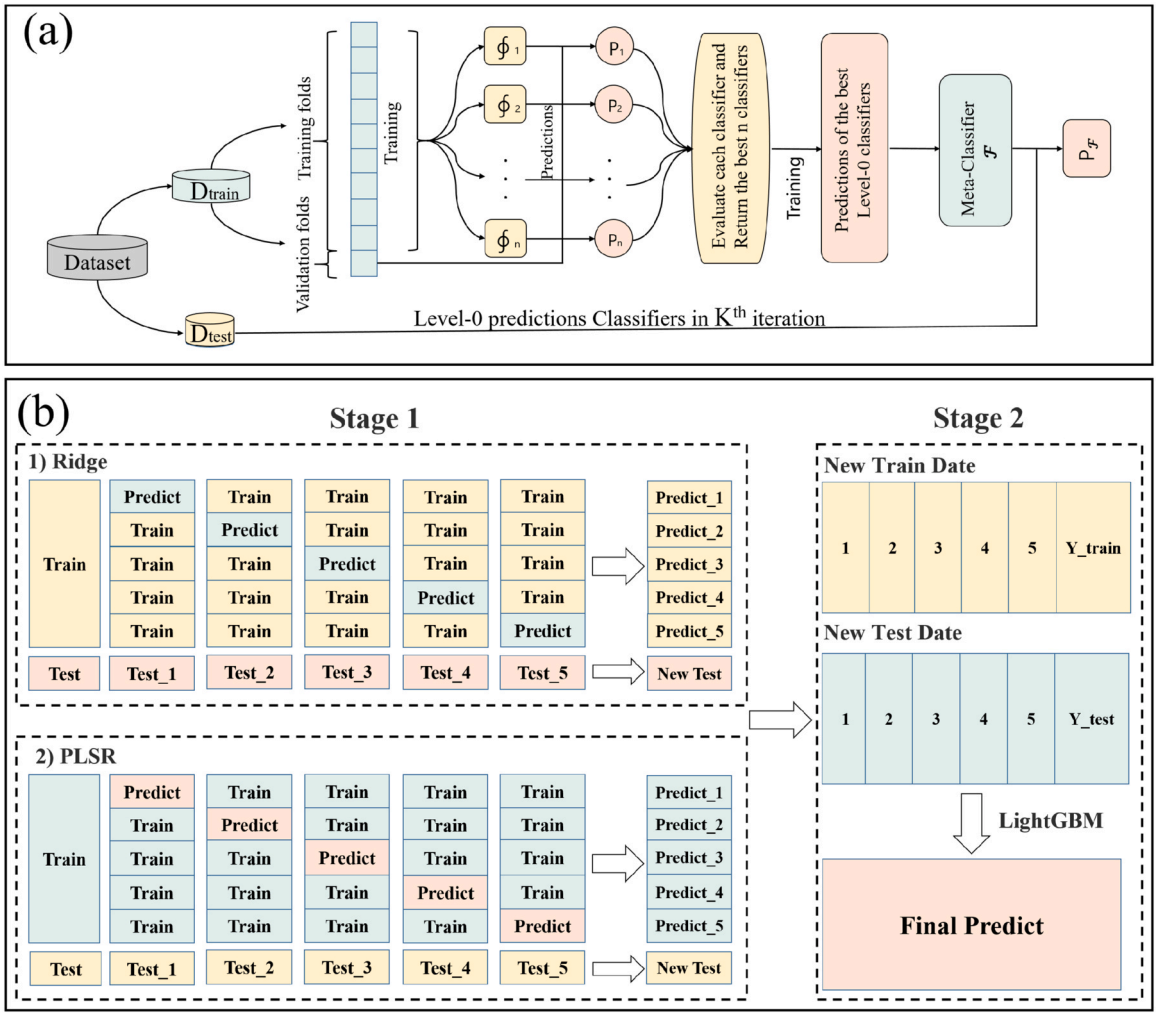


Fig. 2. Principle diagram of integrated learning model algorithm. (a) Schematic diagram of the iterative process of the "integrated learning" method. (b) A multi-layer learning system consisting of Ridge, PLSR and LightGBM.

provide strong support for adulteration concentration regression analysis using chemometrics.

3.2. Adulteration concentration regression analysis based on full band

In the regression analysis, the spectral curve data of the sample was divided into a training set and a test set in a ratio of 0.75:0.25, and random_state=2020 was set to ensure the repeatability of the results. For the training set, the K-fold cross-validation method was used. Experimentation with various algorithms was conducted, including Ridge, LightGBM, CatBoost, RF, Gradient Boosting Decision Tree (GBDT), PLSR algorithm and Stacking model. These algorithms were used to model the preprocessed data. Taking the model built using the LightGBM algorithm as an example, the data model after preprocessing by different methods is modeled. The results are shown in Table 2.

Among various preprocessing methods, the best performance is the data processed by MF. This method yielded significant improvements in both the test set and cross-validation metrics. Specifically, on the test set, its R^2 is 0.857 and RMSE is 0.106; in cross-validation, R^2_{cv} is 0.815 and $RMSE_{cv}$ value is 0.111. It shows that compared with the original spectral reflectance data, the regression results are significantly improved after MF preprocessing. The application of MF can effectively eliminate sudden high-frequency noise, which is beneficial to the smooth processing of data and has a positive impact on subsequent analysis. On the contrary, other preprocessing methods showed varying degrees of reduced or slightly improved effects in subsequent regression analyses (Oboué et al., 2024). Therefore, further research will select MF preprocessed data for analysis to ensure data quality and reliability of results.

The results of the predictive models using MF-processed spectral data are summarized in Table 3. When using full-band information for modeling, it can be seen that the R^2 and R^2_{cv} of the three models using CatBoost, RF and GBDT are both lower than 0.8, which is not suitable for subsequent regression analysis. The R^2 and R^2_{cv} of the three algorithm models of Ridge, LightGBM and PLSR are both higher than 0.8, which is suitable for subsequent regression analysis. The highest R^2 of the single-model test set is the Ridge model, which is 0.930. Its cross-validation R^2_{cv} is 0.852, and the model takes a relatively short time. The study further employs a Stacking model, a type of ensemble learning method that combines predictions from multiple base models with a meta-model (Zhao and Cheng, 2022). The Stacking designed in this study uses Ridge and PLSR as the basic model and LightGBM as the meta-model. The R^2 of the test set increased to 0.943, and the RMSE decreased to 0.066; the cross-validation results were also more ideal, with R^2_{cv} of 0.881 and $RMSE_{cv}$ of 0.089. Compared with the results of three single models such as Ridge, the Stacking model fully combines the advantages of the three models and has achieved significant improvements in all aspects.

After applying the MF preprocessing algorithm, the model was established, which significantly improved the performance of the four

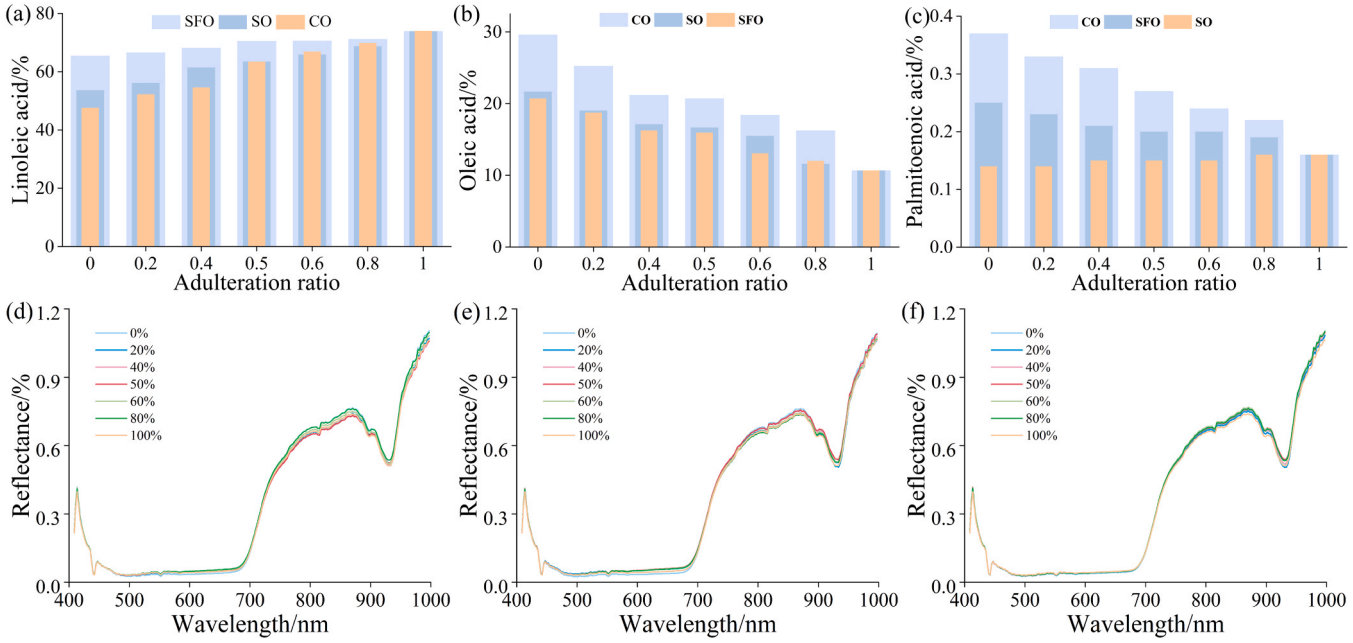


Fig. 3. Fatty acid content and spectral curves of mixed oil samples at different adulteration concentrations. (a) - (c) respectively represent the LA, OA, and PA contents of oil samples under different adulteration concentrations, (d) - (f) respectively represent the spectral curves of SFO, SO, and CO under different adulteration ratios.

Table 2
Results of using LightGBM to model spectral data after different preprocessing methods.

Methods	RMSE	R^2	$RMSE_{cv}$	R^2_{cv}	Fitting time /s
Raw	0.194	0.516	0.190	0.458	3.630
L2NN	0.205	0.462	0.219	0.279	3.993
MSC	0.189	0.543	0.200	0.400	4.117
MF	0.106	0.857	0.111	0.815	3.711

L2NN – L2 norm normalization; MSC – multiple scattering correction; MF – median filtering; RMSE – root mean square error; R^2 – coefficient of determination; $RMSE_{cv}$ – cross-validation root mean square error; R^2_{cv} – cross-validation coefficient of determination.

Table 3
Prediction results of different algorithm models after using MF.

Models	RMSE	R^2	$RMSE_{cv}$	R^2_{cv}	Fitting time /s
Ridge	0.074	0.930	0.099	0.852	0.051
LightGBM	0.106	0.857	0.111	0.815	3.711
CatBoost	0.134	0.769	0.128	0.753	14.636
RF	0.160	0.671	0.153	0.647	0.870
GBDT	0.149	0.717	0.137	0.718	0.332
PLSR	0.083	0.911	0.108	0.826	0.025
Stacking	0.066	0.943	0.089	0.881	4.215

Ridge – ridge regression; RF – random forest; GBDT – gradient boosting decision trees; PLSR – partial least squares regression; RMSE – root mean square error; R^2 – coefficient of determination; $RMSE_{cv}$ – cross-validation root mean square error; R^2_{cv} – cross-validation coefficient of determination.

algorithms. The prediction results of the Ridge, LightGBM and PLSR algorithm models and the Stacking model on SSO adulteration were drawn, as shown in Fig. 4.

3.3. Feature band extraction

In order to reduce complexity during the modeling process and achieve efficient detection, a characteristic band modeling method was adopted. This method used an algorithm model to select band features that were of significant value to the model and eliminate data that was unhelpful or had a small impact on classification to reduce the interference of redundant information on the accuracy of the model. In this way, not only the calculation amount of the model was reduced, but also the goal of rapid detection was achieved. The Ridge model, chosen for its effectiveness in mitigating overfitting and addressing multicollinearity issues through L2 regularization (Zhang et al., 2024). This time, the Ridge model was used to extract the first thirty characteristic bands with weights, and their distribution resulted on the spectrum are shown in Fig. 5a. The selected characteristic bands were distributed near 440, 530 nm and 880 nm – 950 nm. The reflection spectrum values in these vicinity may be related to unsaturated fatty acid. Specifically, the characteristic bands at 430 nm and 530 nm may be affected by riboflavin and vitamin E derivatives, and these bands were associated with carotenoids and chlorophyll regions (Mburu et al., 2021). In addition, the characteristic band near 950 nm may correspond to the second overtone of O-H stretching (Kucha et al., 2022; Lintvedt et al., 2023). Subsequently, only these selected characteristic bands were used for modeling analysis to reduce redundant information in the data, save computing time, and capture the significant features of the spectrum more effectively.

Table 4
Prediction results of spectral data after feature extraction for Ridge using different models.

Methods	Models	RMSE	R^2	$RMSE_{cv}$	R^2_{cv}	Fitting time /s
Ridge	**Ridge**	0.080	0.916	0.077	0.910	0.006
	LightGBM	0.108	0.848	0.105	0.831	2.055
	PLSR	0.081	0.916	0.078	0.907	0.005
	Stacking	0.065	0.944	0.076	0.913	1.991

Ridge – ridge regression; PLSR – partial least squares regression; RMSE – root mean square error; R^2 – coefficient of determination; $RMSE_{cv}$ – cross-validation root mean square error; R^2_{cv} – cross-validation coefficient of determination.

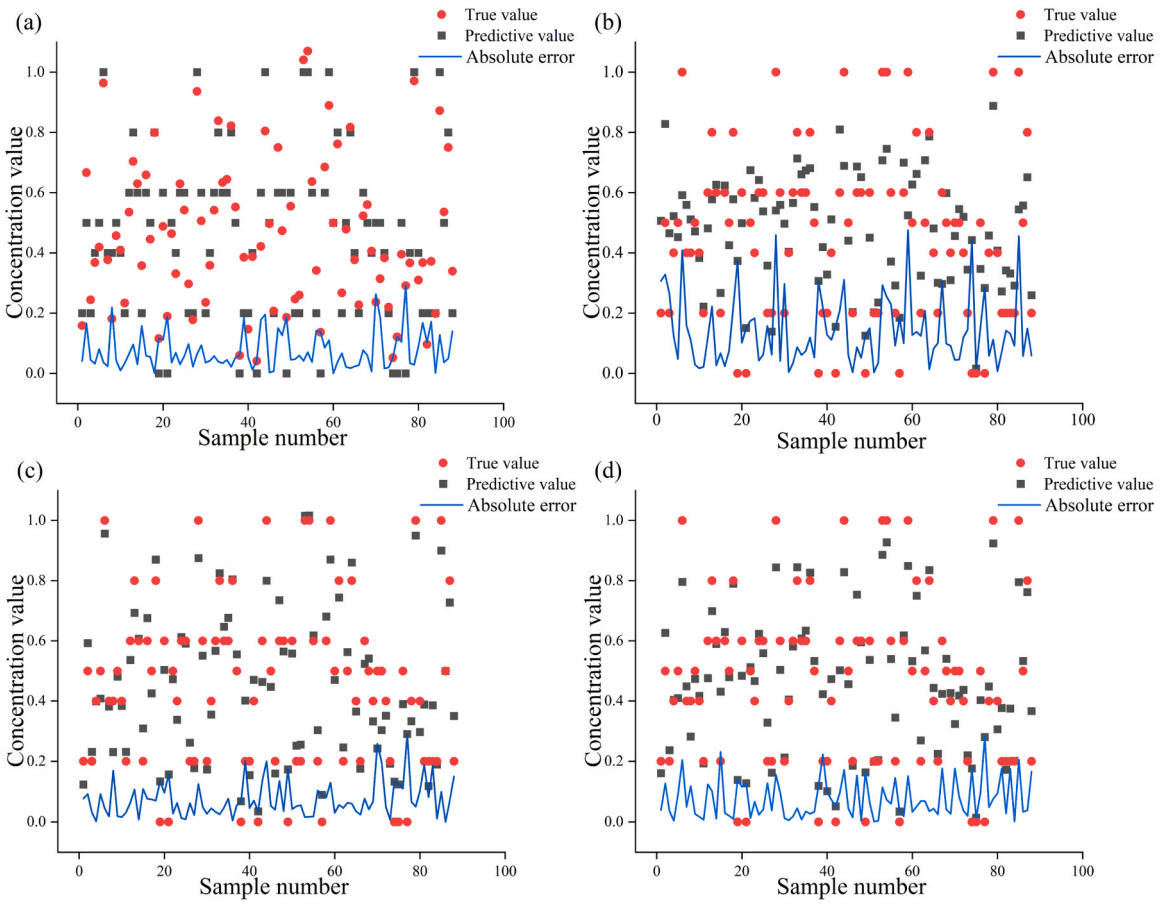


Fig. 4. Modeling results of different algorithm models after using MF preprocessing. (a) Ridge's modeling results. (b) Modeling results of LightGBM. (c) PLSR modeling results. (d) Modeling results of Stacking.

3.4. Regression analysis of adulteration concentration based on characteristic bands

A reanalysis was conducted by applying MF to spectral data and extracting the first 30 feature bands using a Ridge model. It resulted in a significant reduction in the number of features and a notable decrease in modeling time. In comparison to the full-spectrum regression analysis, R^2_{cv} showed varying degrees of improvement, with R^2_{cv} values of 0.910, 0.831, 0.907, and 0.913 for Ridge, LightGBM, PLSR, and Stacking, respectively. Comparative analysis of the experimental results indicated that the modeling performance of the MF-Ridge-Stacking model surpassed that of MF-LightGBM, MF-CatBoost, MF-RF, MF-GBDT, MF-PLSR, MF-Ridge, and MF-stacking. Ultimately, MF-Ridge-Stacking was determined to be the optimal regression analysis model, with R^2_{cv} of 0.913, $RMSE_{cv}$ of 0.076, R^2 of 0.944, and RMSE of 0.065. This series of analyses demonstrates the superiority of regression analysis for adulterant concentration based on feature bands.

3.5. Correlation analysis and deep modeling analysis of the spectral data and fatty acid content

In the study, GC-MS technology was utilized to accurately detect the LA, OA, and PA contents of each sample to ensure the label accuracy of the experimental results. By comprehensively analyzing these fatty acid content and spectral data, an in-depth study of the relationship between the content of each fatty acid and the concentration of SSO was conducted to further verify the scientific rationality of the experiment. Finally, detailed data presenting the levels of LA, OA, and PA in the nineteen oil samples are provided in Table S1.

To conduct an in-depth analysis of the relationship between each characteristic and the content of each fatty acid, data from oil samples with different doping concentrations was integrated. The Ridge model was used to extract the first thirty characteristic bands, and the relationship between LA, OA, PA and doping concentration was further explored. The results show that there is a strong correlation between the doping concentration and the LA content (Fig. 6b), with a correlation coefficient r value of 0.75. In specific wavebands (525 nm, 552 nm and 609 nm), the spectral data shows a positive correlation with the doping concentration (Fig. 6a), with a partial correlation coefficient p<=0.001. There is a strong negative correlation between LA and OA, with a correlation coefficient r of −0.87, that is, when the LA content is higher, the OA content is also higher (Yeilaghi et al., 2012). Furthermore, three fatty acid contents and the spectral data after preprocessing (MF) were included into a multivariate regression model to explore the intrinsic relationship between them. The model analysis results are shown in Table 5. It can be clearly seen that after adding the three fatty acid contents as features to model training, the overall performance has been improved. The R^2 value of each model is 0.9 or above, especially the R^2 of the Stacking model reaches 0.976, indicating that this comprehensive model shows excellent performance in prediction.

The Stacking algorithm was used to test the oil sample data. The test samples were 88 additional samples with adulteration concentrations ranging from 0 % to 100 %. The overall absolute errors are relatively small, with the maximum deviation being 16.64 % and the minimum deviation being 0.01 %. The results show that Stacking has a good prediction effect on adulteration in different oil samples. The model has a good prediction effect on the adulteration of different samples, which verifies its reliability and effectiveness in practical applications and provides strong support for authenticity testing in the food industry.

GC-MS technology has proven effective in prior studies for both

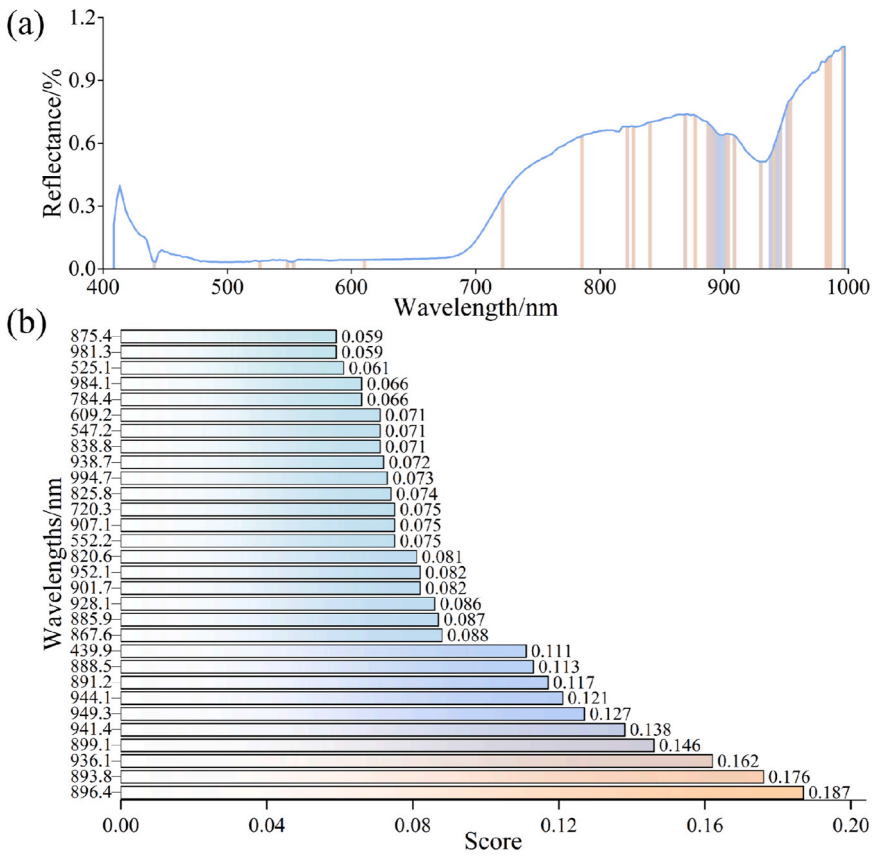


Fig. 5. Ridge feature extraction diagram. (a) Characteristic wavelength distribution diagram selected by Ridge. (b) Ranking of scores of the first 30 feature wavelengths extracted by Ridge.

quantitative and qualitative analysis of oil authenticity (Syafri et al., 2022). In the current study, many researchers have utilized GC-MS for comparative quantitative analysis of oil (Dinkar Gore et al., 2023). However, methods for identifying and validating specific oil types using single fatty acid are relatively simple and not reliable enough when samples vary (Sales et al., 2019). In contrast, it is more sensitive to identify the relative proportions of LA, OA, PA and other components in oil samples (X. Wang et al., 2021). GC-MS technology was used to analyze a variety of fatty acid, combined with chemometrics to determine the geographical origin of extra virgin olive oil, and successfully classified regional categories accurately through partial least squares discriminant analysis (Peršurić et al., 2018). GC-MS technology was combined with chemometric methods to detect fraud in sesame oil and proposed a new detection method (Aghili et al., 2022). Although GC-MS technology has shown advantages in multiple studies, it still has certain limitations in its stability and cannot provide consistent results for certain compounds with volatile components (Rifna et al., 2022). In addition, when oils are mixed and adulterated with multiple components, traditional chromatographic techniques cannot effectively distinguish (Peng et al., 2015). Nevertheless, GC-MS delivers high-resolution chemical insights by analyzing a spectrum of fatty acids. HSI technology, with its high sensitivity to molecular vibration information, provides unique insights into the structural information of different functional groups in SSO. Therefore, the combination of chromatographic technology and spectroscopic technology can more fully reflect the complexity of the sample, which may create unexpected effects.

4. Conclusion

This study proposes a chemometric method for detecting SSO adulteration concentration based on HSI and GC-MS technologies. Through different preprocessing, it was found that using MF for preprocessing can successfully reduce noise information and significantly improve the robustness and generalization ability of the model. The study also identifies specific wavelength bands (near 440 nm, 530 nm, and 880 nm – 950 nm) that optimize modeling time without compromising prediction accuracy. When developing the model, the constructed integrated learning model (Ridge, PLSR basic model, LightGBM meta-model) showed higher accuracy than a single model in predicting SSO concentration adulteration. In particular, by jointly modeling the fatty acid content measured by GC-MS and hyperspectral data, R^2 was increased to 0.976, further highlighting the superior performance of the model. Therefore, MF-Ridge-Stacking was determined to be the best model for adulteration concentration prediction of SSO. This study expanded the identification method of adulteration of SSO and other edible oils, and provided a practical and reliable adulteration detection method for the food industry by combining GC-MS and HSI technology, as well as machine learning. Future research can consider further expanding the sample range, using advanced instrument technology, and exploring other spectral denoising methods in more depth to improve the robustness of the model. The findings can help increase consumer confidence in food authenticity and create a healthier, more sustainable market environment. Additionally, the model's performance at low adulteration levels (such as 1 %, 2 %) should be tested in future to ensure it can accurately and effectively operate across various adulteration levels.

CRediT authorship contribution statement

Lijia Xu: Funding acquisition. **Shutao Yin:** Funding acquisition. **Qianlong Wang:** Writing – review & editing, Methodology,

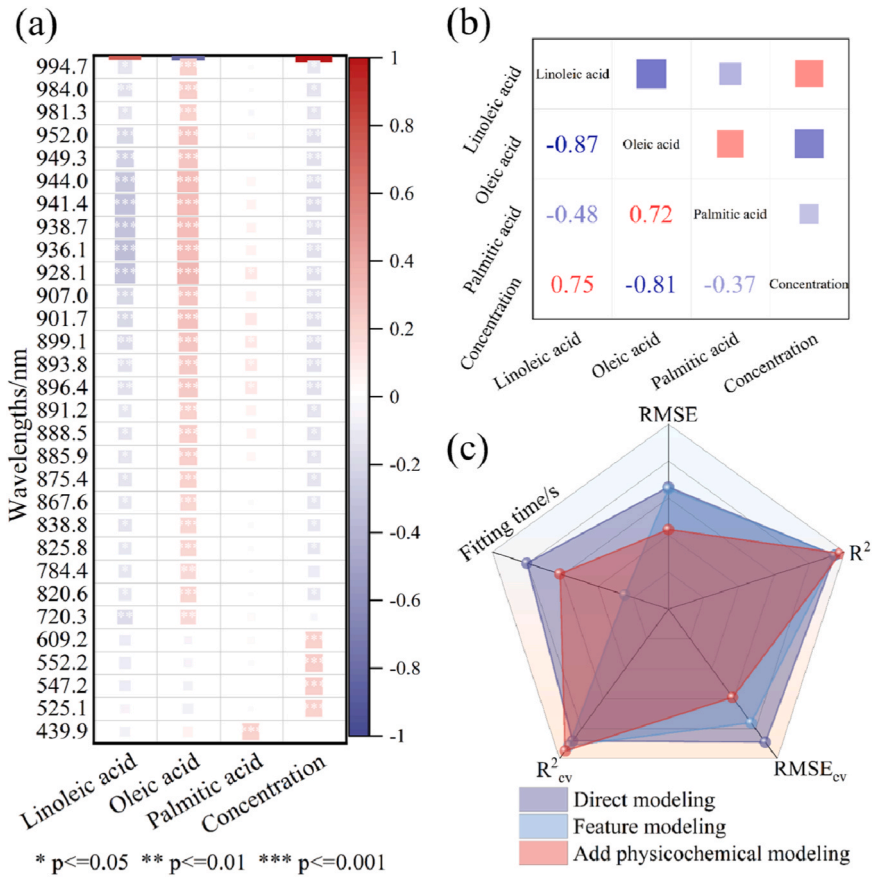


Fig. 6. Spectral characteristic band analysis diagram combined with different concentrations of fatty acids. (a) Correlation analysis between spectral characteristic bands and fatty acid contents at different adulteration concentrations. (b) orrelation analysis between adulteration concentration and fatty acid content. (c) Radar chart of performance comparison of different models.

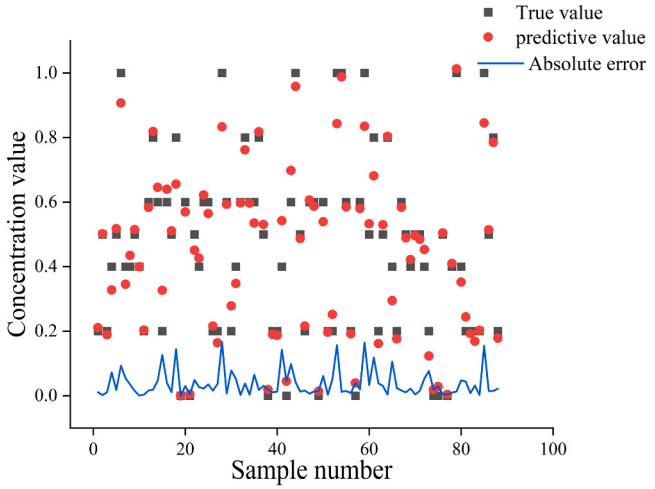


Fig. 7. Stacking prediction result graph.

Conceptualization. **Zhiyong Zou:** Conceptualization. **Meiliang Li:** Project administration. **Qiang Cui:** Formal analysis. **Man Zhou:** Formal analysis, Data curation. **Jiangbo Zhen:** Writing – original draft. **Menghua Li:** Writing – review & editing, Methodology. **Qing Zhang:** Data curation. **Dongyu Yuan:** Writing – original draft.

Declaration of Competing Interest

We declare that we have neither financial nor personal relationships with other people or organizations that can inappropriately influence our work; there is no professional or any personal interest of any nature or kind in any product, service or company that could be considered as influencing the position or the review presented in the manuscript entitled "Combination of gas chromatography-mass spectrometry and hyperspectral imaging for identification of adulterated Safflower seed oil".

Table 5
Results of modeling combining fatty acid content and full-band spectral data.

Models	RMSE	R^2	$RMSE_{cv}$	R^2_{cv}	Fitting time /s
Ridge	0.068	0.940	0.083	0.896	0.042
LightGBM	0.049	0.969	0.056	0.954	3.544
CatBoost	0.059	0.955	0.061	0.944	13.948
RF	0.089	0.900	0.079	0.906	0.691
GBDT	0.078	0.922	0.069	0.929	0.211
PLSR	0.082	0.915	0.087	0.887	0.018
Stacking	0.043	0.976	0.059	0.948	3.465

Ridge – ridge regression; RF – random forest; GBDT – gradient boosting decision trees; PLSR – partial least squares regression; RMSE – root mean square error; R^2 – coefficient of determination; $RMSE_{cv}$ – cross-validation root mean square error; R^2_{cv} – cross-validation coefficient of determination.

Data availability

Data will be made available on request.

Acknowledgements

This work is supported by CAU-SCCD Advanced Agricultural & Industrial Institute (SCB-ZNCY-2022005) and Sichuan Province Science and Technology Support Program (Grant No. 2022NZZJ0034).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jfca.2024.106593.

References

Abellán, J., Mantas, C.J., Castellano, J.G., Moral-García, S., 2018. Increasing diversity in random forest learning algorithm via imprecise probabilities. Expert Syst. Appl. *97*, 228–243. https://doi.org/10.1016/j.eswa.2017.12.029.

Aghili, N.S., Rasekh, M., Karami, H., Azizi, V., Gancarz, M., 2022. Detection of fraud in sesame oil with the help of artificial intelligence combined with chemometrics methods and chemical compounds characterization by gas chromatography–mass spectrometry. LWT *167*, 113863. https://doi.org/10.1016/j.lwt.2022.113863.

Appiah, O., Asante, M., Hayfron-Acquah, J.B., 2022. Improved approximated median filter algorithm for real-time computer vision applications. J. King Saud. Univ. - Comput. Inf. Sci. *34* (3), 782–792. https://doi.org/10.1016/j.jksuci.2020.04.005.

Aulia, R., Kim, Y., Zuhrotul Amanah, H., Muhammad Akbar Andi, A., Kim, H., Kim, H., Cho, B.-K., 2022. Non-destructive prediction of protein contents of soybean seeds using near-infrared hyperspectral imaging. Infrared Phys. Technol. *127*, 104365 https://doi.org/10.1016/j.infrared.2022.104365.

Bao, R., Tang, F., Rich, C., Hatzakis, E., 2023. A comparative evaluation of low-field and high-field NMR untargeted analysis: authentication of virgin coconut oil adulterated with refined coconut oil as a case study. Anal. Chim. Acta *1273*, 341537. https://doi.org/10.1016/j.aca.2023.341537.

Cao, G., Hong, Y., Wu, H., Chen, Z., Lu, M., Cai, Z., 2021. Visual authentication of edible vegetable oil and used cooking oil using MALDI imaging mass spectrometry. Food Control *125*, 107966. https://doi.org/10.1016/j.foodcont.2021.107966.

Chen, C., Zhang, Q.M., Ma, Q., Yu, B., 2019. LightGBM-PPI: Predicting protein-protein interactions through LightGBM with multi-information fusion. Chemom. Intell. Lab. Syst. *191*, 54–64. https://doi.org/10.1016/j.chemolab.2019.06.003.

Contreras, M. d M., Arroyo-Manzanares, N., Arce, C., Arce, L., 2019. HS-GC-IMS and chemometric data treatment for food authenticity assessment: Olive oil mapping and classification through two different devices as an example. Food Control *98*, 82–93. https://doi.org/10.1016/j.foodcont.2018.11.001.

Cruz-Tirado, J.P., Muñoz-Pastor, D., de Moraes, I.A., Lima, A.F., Godoy, H.T., Barbin, D. F., Siche, R., 2023. Comparing data driven soft independent class analogy (DD-SIMCA) and one class partial least square (OC-PLS) to authenticate sacha inchi (Plukenetia volubilis L.) oil using portable NIR spectrometer. Chemom. Intell. Lab. Syst. *242*, 105004 https://doi.org/10.1016/j.chemolab.2023.105004.

Cui, R., Li, J., Wang, Y., Fang, S., Yu, K., Zhao, Y., 2022. Hyperspectral imaging coupled with Dual-channel convolutional neural network for early detection of apple valsa canker. Comput. Electron. Agric. *202*, 107411 https://doi.org/10.1016/j.compag.2022.107411.

Ding, Y., Chen, Z.Q., Lu, W.F., Wang, X.Q., 2021. A CatBoost approach with wavelet decomposition to improve satellite-derived high-resolution PM2.5 estimates in Beijing-Tianjin-Hebei. Atmos. Environ. *249*, 118212 https://doi.org/10.1016/j.atmosenv.2021.118212.

Dinkar Gore, D., Ahmad, F., Tikoo, K., Kumar Bansal, A., Kumar, D., Pal Singh, I., 2023. Comparative quantitative analysis of fruit oil from Hippophae rhamnoides (seabuckthorn) by qNMR, FTIR and GC–MS. Chin. Herb. Med. *15* (4), 607–613. https://doi.org/10.1016/j.chmed.2023.05.005.

Fadiloglu, E.E., Yildiz Turp, G., Celebioglu, C., Sel, G., 2023. Influence of different cooking methods on quality characteristics and nutritional value of gluten-free beef burger patties formulated with walnut oil, safflower oil and buckwheat. Meat Sci. *204*, 109251 https://doi.org/10.1016/j.meatsci.2023.109251.

Fujioka, K., Sun, R., 2022. Interpolating Moving Ridge Regression (IMRR): A machine learning algorithm to predict energy gradients for ab initio molecular dynamics simulations. Chem. Phys. *557*, 111482 https://doi.org/10.1016/j.chemphys.2022.111482.

Gerhardt, N., Schwolow, S., Rohn, S., Pérez-Cacho, P.R., Galán-Soldevilla, H., Arce, L., Weller, P., 2019. Quality assessment of olive oils based on temperature-ramped HS-GC-IMS and sensory evaluation: comparison of different processing approaches by LDA, kNN, and SVM. Food Chem. *278*, 720–728. https://doi.org/10.1016/j.foodchem.2018.11.095.

Han, L., Chen, M., Li, Y., Wu, S., Zhang, L., Tu, K., Song, L., 2022. Discrimination of different oil types and adulterated safflower seed oil based on electronic nose combined with gas chromatography-ion mobility spectrometry. J. Food Compos. Anal. *114*, 104804 https://doi.org/10.1016/j.jfca.2022.104804.

Hertzog, J., Garnier, C., Mase, C., Mariette, S., Serve, O., Hubert-Roux, M., Barrère-Mangote, C., 2022. Fractionation by flash chromatography and molecular characterization of bio-oil by ultra-high-resolution mass spectrometry and NMR spectroscopy. J. Anal. Appl. Pyrolysis *166*, 105611. https://doi.org/10.1016/j.jaap.2022.105611.

Hou, G., Liu, S., Zhu, Y., Yu, S., Zha, Z., Zhao, Q., Chu, J., 2024. Fabrication of Ag/PP terahertz hollow waveguide by vacuum evaporation method and transmission performance investigations. Opt. Mater. *147*, 114710 https://doi.org/10.1016/j.optmat.2023.114710.

Jermwongruttanachai, P., Pathaveerat, S., Noypitak, S., 2023. Quantification of adulteration concentration of palm kernel oil in virgin coconut oil using near-infrared hyperspectral imaging1. J. Integr. Agric. https://doi.org/10.1016/j.jia.2023.08.002.

Jin, H., Li, H., Yin, Z., Zhu, Y., Lu, A., Zhao, D., Li, C., 2021. Application of Raman spectroscopy in the rapid detection of waste cooking oil. Food Chem. *362*, 130191 https://doi.org/10.1016/j.foodchem.2021.130191.

Johnson, J.B., Thani, P.R., Naiker, M., 2022. Detection of eucalyptus oil adulteration in Australian tea tree oil using UV–Vis and fluorescence spectroscopy. Talanta Open *6*, 100169. https://doi.org/10.1016/j.talo.2022.100169.

Jurado-Campos, N., Arroyo-Manzanares, N., Viñas, P., Arce, L., 2020. Quality authentication of virgin olive oils using orthogonal techniques and chemometrics based on individual and high-level data fusion information. Talanta *219*, 121260. https://doi.org/10.1016/j.talanta.2020.121260.

Kammili, A., Yadav, P., 2022. Enhancing oleic acid and oil content in low oil and oleic type Indian safflower (Carthamus tinctorius L.). Ind. Crops Prod. *175*, 114254 https://doi.org/10.1016/j.indcrop.2021.114254.

Kanny, G., Fremont, S., Nicolas, J.P., Moneret-Vautrin, D.A., 1994. Food allergy to sunflower oil in a patient sensitized to mugwort pollen. Allergy *49* (7), 561–564. https://doi.org/10.1111/j.1398-9995.1994.tb01129.x.

Khalid, N., Khan, R.S., Hussain, M.I., Farooq, M., Ahmad, A., Ahmed, I., 2017. A comprehensive characterisation of safflower oil for its potential applications as a bioactive food ingredient - a review. Trends Food Sci. Technol. *66*, 176–186. https://doi.org/10.1016/j.tifs.2017.06.009.

Kucha, C.T., Liu, L., Ngadi, M., Gariépy, C., 2022. Prediction and visualization of fat content in polythene-packed meat using near-infrared hyperspectral imaging and chemometrics. J. Food Compos. Anal. *111*, 104633 https://doi.org/10.1016/j.jfca.2022.104633.

Lewis, S.W., 2023. Spectroscopic Techniques. In: Houck, M.M. (Ed.), Encyclopedia of Forensic Sciences, Third Edition. Elsevier, pp. 573–582. https://doi.org/10.1016/B978-0-12-823677-2.00078-7.

Lintvedt, T.A., Andersen, P.V., Afseth, N.K., Heia, K., Lindberg, S.-K., Wold, J.P., 2023. Raman spectroscopy and NIR hyperspectral imaging for in-line estimation of fatty acid features in salmon fillets. Talanta *254*, 124113. https://doi.org/10.1016/j.talanta.2022.124113.

Liu, S., Zeng, A., Lau, K., Ren, C., Chan, P.W., Ng, E., 2021. Predicting long-term monthly electricity demand under future climatic and socioeconomic changes using data-driven methods: a case study of Hong Kong. Sustain. Cities Soc. *70*, 102936 https://doi.org/10.1016/j.scs.2021.102936.

Liu, Y., Li, Q., Yang, W., Sun, B., Zhou, Y., Zheng, Y., Yang, W., 2020. Characterization of the potent odorants in Zanthoxylum armatum DC Prodr. pericarp oil by application of gas chromatography–mass spectrometry–olfactometry and odor activity value. Food Chem. *319*, 126564 https://doi.org/10.1016/j.foodchem.2020.126564.

Mburu, M., Komu, C., Paquet-Durand, O., Hitzmann, B., Zettel, V., 2021. Chia Oil Adulteration Detection Based on Spectroscopic Measurements. Foods *10* (8), 1798. https://doi.org/10.3390/foods10081798.

Meng, X., Yin, C., Yuan, L., Zhang, Y., Ju, Y., Xin, K., Hu, L., 2023. Rapid detection of adulteration of olive oil with soybean oil combined with chemometrics by Fourier transform infrared, visible-near-infrared and excitation-emission matrix fluorescence spectroscopy: a comparative study. Food Chem. *405*, 134828 https://doi.org/10.1016/j.foodchem.2022.134828.

Oboué, Y.A.S.I., Chen, Y., Fomel, S., Zhong, W., Chen, Y., 2024. An advanced median filter for improving the signal-to-noise ratio of seismological datasets. Comput. Geosci. *182*, 105464 https://doi.org/10.1016/j.cageo.2023.105464.

Pan, Z., Li, R.H., Cui, Y.Y., Wu, X.J., Zhang, Y.Y., Wang, Y.T., 2021. A simple and quick method to detect adulterated sesame oil using 3D fluorescence spectra. Spectrochim. Acta Part A: Mol. Biomol. Spectrosc. *245*, 118948 https://doi.org/10.1016/j.saa.2020.118948.

Peng, D., Bi, Y., Ren, X., Yang, G., Sun, S., Wang, X., 2015. Detection and quantification of adulteration of sesame oils with vegetable oils using gas chromatography and multivariate data analysis. Food Chem. *188*, 415–421. https://doi.org/10.1016/j.foodchem.2015.05.001.

Peršurić, Ž., Saftić, L., Mašek, T., Kraljević Pavelić, S., 2018. Comparison of triacylglycerol analysis by MALDI-TOF/MS, fatty acid analysis by GC-MS and non-selective analysis by NIRS in combination with chemometrics for determination of extra virgin olive oil geographical origin. A case study. LWT *95*, 326–332. https://doi.org/10.1016/j.lwt.2018.04.072.

Rifna, E.J., Pandiselvam, R., Kothakota, A., Subba Rao, K.V., Dwivedi, M., Kumar, M., Ramesh, S.V., 2022. Advanced process analytical tools for identification of adulterants in edible oils – a review. Food Chem. *369*, 130898 https://doi.org/10.1016/j.foodchem.2021.130898.

Ríos-Reina, R., Salatti-Dorado, J.Á., Ortiz-Romero, C., Cardador, M.J., Arce, L., Callejón, R., 2024. A comparative study of fluorescence and Raman spectroscopy for discrimination of virgin olive oil categories: chemometric approaches and evaluation against other techniques. Food Control *158*, 110250. https://doi.org/10.1016/j.foodcont.2023.110250.

Rudolphi, S., Becker, H.C., Schierholt, A., von Witzke-Ehbrecht, S., 2012. Improved Estimation of Oil, Linoleic and Oleic Acid and Seed Hull Fractions in Safflower by NIRS. J. Am. OIL Chem. Soc. *89* (3), 363–369. https://doi.org/10.1007/s11746-011-1920-y.

Saha, D., Senthilkumar, T., Singh, C.B., Manickavasagan, A., 2023. Quantitative detection of metanil yellow adulteration in chickpea flour using line-scan near-infrared hyperspectral imaging with partial least square regression and one-

dimensional convolutional neural network. J. Food Compos. Anal., 105290 https://doi.org/10.1016/j.jfca.2023.105290.
Sales, C., Portolés, T., Johnsen, L.G., Danielsen, M., Beltran, J., 2019. Olive oil quality classification and measurement of its organoleptic attributes by untargeted GC–MS and multivariate statistical-based approach. Food Chem. *271*, 488–496. https://doi.org/10.1016/j.foodchem.2018.07.200.
Spadafora, N.D., Mascrez, S., McGregor, L., Purcaro, G., 2022. Exploring multiple-cumulative trapping solid-phase microextraction coupled to gas chromatography–mass spectrometry for quality and authenticity assessment of olive oil. Food Chem. *383*, 132438 https://doi.org/10.1016/j.foodchem.2022.132438.
Spencer, G.F., Herb, S.F., Gormisky, P.J., 1976. Fatty acid composition as a basis for identification of commercial fats and oils. J. Am. Oil Chem. Soc. *53* (3), 94–96. https://doi.org/10.1007/BF02635956.
Syafri, S., Jaswir, I., Yusof, F., Rohman, A., Ahda, M., Hamidi, D., 2022. The use of instrumental technique and chemometrics for essential oil authentication: A review. Results Chem. *4*, 100622 https://doi.org/10.1016/j.rechem.2022.100622.
Tian, P., Meng, Q., Wu, Z., Lin, J., Huang, X., Zhu, H., Li, Y., 2023. Detection of mango soluble solid content using hyperspectral imaging technology. Infrared Phys. Technol. *129*, 104576 https://doi.org/10.1016/j.infrared.2023.104576.
Uncu, O., Ozen, B., Tokatli, F., 2019. Mid-infrared spectroscopic detection of sunflower oil adulteration with safflower oil. GRASAS Y ACEITES *70* (1), e290. https://doi.org/10.3989/gya.0579181.
Wang, F., Ma, S., Wang, H., Li, Y., Qin, Z., Zhang, J., 2018. A hybrid model integrating improved flower pollination algorithm-based feature selection and improved random forest for NOX emission estimation of coal-fired power plants. Measurement *125*, 303–312. https://doi.org/10.1016/j.measurement.2018.04.069.
Wang, S.-L., Zhu, R.-Y., Zhang, X.-L., Shen, W.-X., Zhang, H., Fang, X.-W., Yang, S.-P., 2021. Rapid screening of low-quality cooking oil by extractive electrospray ionization mass spectrometry. Chin. J. Anal. Chem. *49* (12), 43–48. https://doi.org/10.1016/j.cjac.2021.09.003.
Wang, X., Han, Y., Li, Y., Wang, S., Wang, J., 2021. Detection of Qinghai-Tibet Plateau flaxseed oil adulteration based on fatty acid profiles and chemometrics. Food Control *130*, 108332. https://doi.org/10.1016/j.foodcont.2021.108332.
Wang, Y., Song, S., 2023. Variety identification of sweet maize seeds based on hyperspectral imaging combined with deep learning. Infrared Phys. Technol. *130*, 104611 https://doi.org/10.1016/j.infrared.2023.104611.
Wei, Q., Liu, G., Zhang, C., Sun, J., Zhang, Y., 2022. Identification of characteristic volatile compounds and prediction of fermentation degree of pomelo wine using partial least squares regression. LWT *154*, 112830. https://doi.org/10.1016/j.lwt.2021.112830.
Yeilaghi, H., Arzani, A., Ghaderian, M., Fotovat, R., Feizi, M., Pourdad, S.S., 2012. Effect of salinity on seed oil content and fatty acid composition of safflower (Carthamus tinctorius L.) genotypes. Food Chem. *130* (3), 618–625. https://doi.org/10.1016/j.foodchem.2011.07.085.
Zhan, H., Wu, S., Bao, R., Ge, L., Zhao, K., 2015. Qualitative identification of crude oils from different oil fields using terahertz time-domain spectroscopy. Fuel *143*, 189–193. https://doi.org/10.1016/j.fuel.2014.11.047.
Zhang, H., Zhao, J., Huang, L., Huang, W., Dong, Y., Ma, H., Ruan, C., 2024. Development of new indices and use of CARS-Ridge algorithm for wheat fusarium head blight detection using in-situ hyperspectral data. Biosyst. Eng. *237*, 13–25. https://doi.org/10.1016/j.biosystemseng.2023.11.009.
Zhang, J.-J., Gao, Y., Zhao, M.-L., Xu, X., Xi, B.-N., Lin, L.-K., Shen, Y., 2023. Detection of walnut oil adulterated with high-linoleic acid vegetable oils using triacylglycerol pseudotargeted method based on SFC-QTOF-MS. Food Chem. *416*, 135837 https://doi.org/10.1016/j.foodchem.2023.135837.
Zhao, A.B., Cheng, T.T., 2022. Stock return prediction: stacking a variety of models. J. Empir. Financ. *67*, 288–317. https://doi.org/10.1016/j.jempfin.2022.04.001.
Zhou, M., Wang, L., Wu, H.J., Li, Q.Y., Li, M.L., Zhang, Z.Q., Zou, Z.Y., 2022. Machine learning modeling and prediction of peanut protein content based on spectral images and stoichiometry. LWT-FOOD Sci. Technol. *169*, 114015 https://doi.org/10.1016/j.lwt.2022.114015.